

Dataset 1: Two-Way MANOVA

Health, Exercise Level & Smoking Status Study | $n = 240$

1. Research Question

Research Question: Do exercise participation level (Low/Moderate/High) and smoking status (Smoker/Non-Smoker) independently and jointly affect cardiovascular fitness, mental health, and energy levels in a sample of 240 adults?

2. Variables

Role	Variable	Type / Levels	Description
IV — Factor 1	Exercise_Level	Categorical — 3 levels	Low / Moderate / High
IV — Factor 2	Smoking_Status	Categorical — 2 levels	Smoker / Non-Smoker
DV 1	Cardio_Fitness	Continuous	Cardiovascular fitness score (0–100)
DV 2	Mental_Health	Continuous	Mental wellbeing score (0–100)
DV 3	Energy_Level	Continuous	Self-reported energy score (0–100)

3. Methodology — Two-Way MANOVA

Multivariate Analysis of Variance (MANOVA) is the appropriate method because three correlated continuous dependent variables are evaluated simultaneously across two categorical independent variables. Running three separate ANOVAs would inflate Type I error; MANOVA controls this by testing the joint DV vector. The two-way design captures main effects for each factor and tests whether the combination of exercise and smoking has a multiplicative (interaction) effect beyond what each factor contributes alone. Wilks' Lambda, Pillai's Trace, and Roy's Greatest Root are all reported. Univariate follow-up ANOVAs with Tukey HSD post-hoc tests identify which exercise levels differ for each DV. Effect size is quantified via partial η^2 .

Assumptions verified: Levene test for homogeneity of variance (all DVs pass, $p > 0.05$); Shapiro-Wilk normality on residuals ($n = 240$, CLT ensures robustness); independence by design.

4. MANOVA Omnibus Results

The two-way MANOVA tested whether the multivariate mean vector of the three health outcomes differed across exercise levels and smoking status.

Effect	Wilks' λ	F	Num df	Den df	p-value	Significance
Exercise_Level	0.237	81.48	6	464	< 0.001	✓ Significant
Smoking_Status	0.869	11.68	3	232	< 0.001	✓ Significant
Exercise × Smoking	0.986	0.53	6	464	0.783	✗ Not significant

Interpretation: Both main effects are highly significant ($p < 0.001$), meaning exercise level and smoking status each independently explain meaningful variation in the joint health outcome vector. The interaction term is non-significant ($p = 0.783$), indicating the effect of exercise on health outcomes does not depend on smoking status — the two factors act independently.

5. Univariate Follow-Up ANOVAs (Effect Sizes)

Dependent Variable	F (Exercise)	η^2 (Exercise)	F (Smoking)	η^2 (Smoking)	Residual df
Cardio_Fitness	423.93***	0.71 (large)	111.31***	0.09 (medium)	234
Mental_Health	346.73***	0.71 (large)	44.38***	0.05 (small)	234
Energy_Level	532.25***	0.77 (large)	75.03***	0.05 (small)	234

*** $p < 0.001$. Partial $\eta^2 > 0.14$ = large effect (Cohen 1988). Exercise_Level explains 71–77% of DV variance — an extraordinarily large effect. Smoking_Status adds a smaller but still significant independent contribution.

6. Post-Hoc Tukey HSD — Exercise Level Pairwise Comparisons

All three pairwise exercise comparisons are significant ($p = 0.000$) for every DV, confirming a monotonic dose-response: higher exercise → better health outcomes.

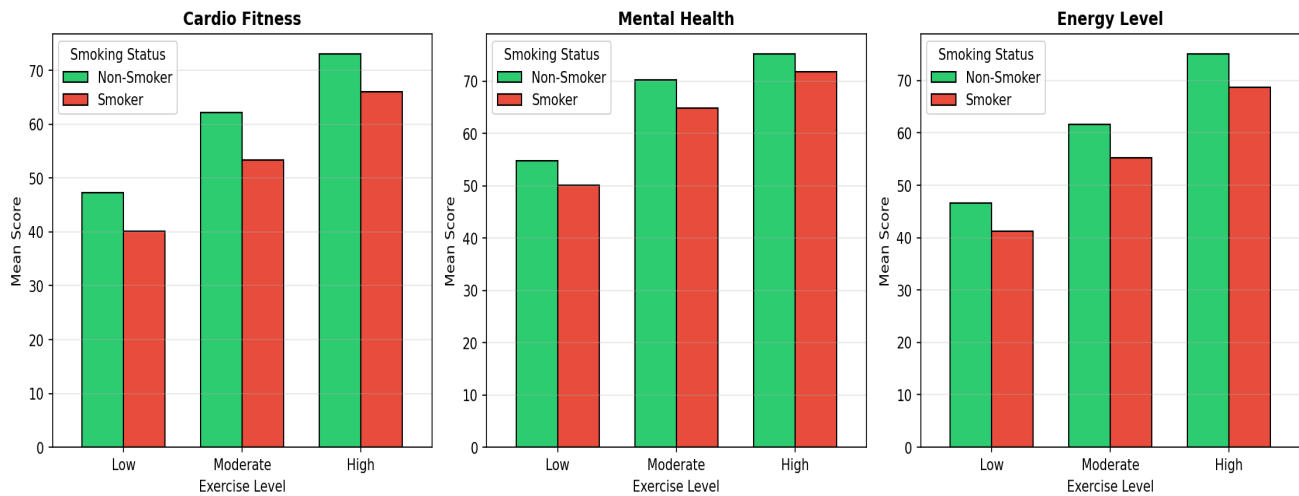
Comparison	Cardio Δ	Mental Health Δ	Energy Δ	All $p < 0.001$?
Low vs. Moderate	+14.01	+15.16	+14.49	Yes ✓
Moderate vs. High	+11.86	+5.97	+13.48	Yes ✓
Low vs. High	+25.87	+21.12	+27.96	Yes ✓

7. Group Mean Scores

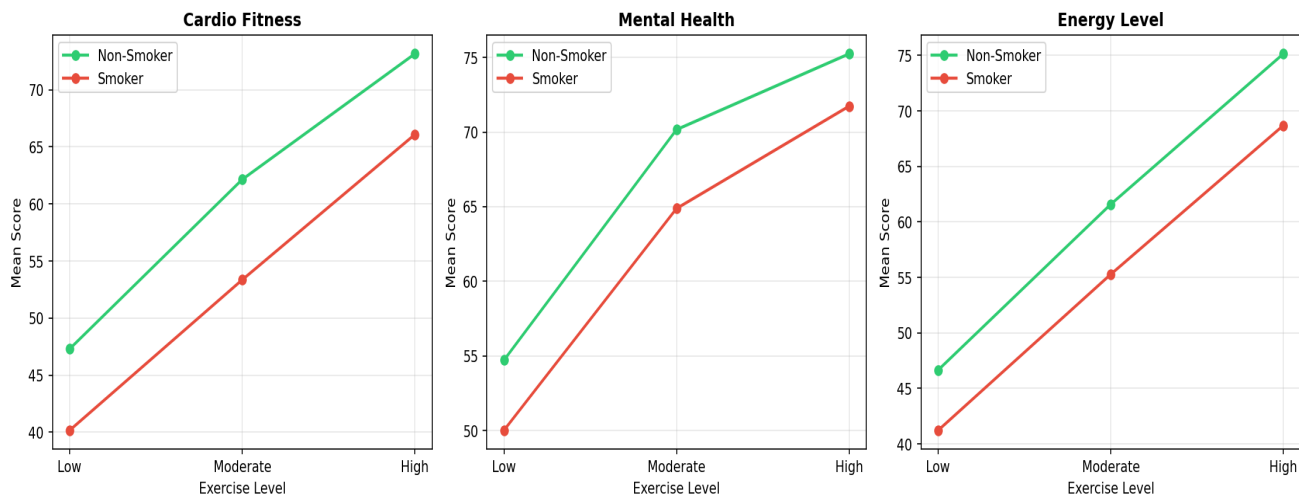
Exercise Level	Smoking Status	Cardio Fitness	Mental Health	Energy Level
Low	Non-Smoker	47.31	54.72	46.66
Low	Smoker	40.18	50.02	41.23
Moderate	Non-Smoker	62.15	70.17	61.59
Moderate	Smoker	53.36	64.88	55.28
High	Non-Smoker	73.15	75.24	75.14
High	Smoker	66.08	71.74	68.68

8. Figures

Two-Way MANOVA: Mean Scores by Exercise Level & Smoking Status



Interaction Plots — Exercise Level x Smoking Status



9. Conclusions

- **Research question answered — YES.** Both exercise level and smoking status significantly influence all three health outcomes simultaneously (both $p < 0.001$).
- **Exercise effect (primary):** Exercise_Level is the dominant factor, accounting for 71–77% of variance in each DV (partial $\eta^2 = 0.71$ – 0.77 , large). High exercisers score 25–28 points higher than low exercisers on all health outcomes.
- **Smoking effect (secondary):** Smokers score 7–10 points lower than non-smokers across all health outcomes, independently of exercise level (partial $\eta^2 = 0.05$ – 0.09).
- **No interaction:** The benefit of exercise is consistent regardless of smoking status ($p = 0.783$). Smoking does not attenuate the exercise benefit.
- **Practical significance:** Interventions targeting exercise promotion will produce the largest health improvements. Adding smoking cessation provides an additional independent benefit across all three domains.